

COURSE DESCRIPTION

Course Number	0418-391	Course Title	Cloud Computing Fundamentals
Credit Hours	3 (3:0:3)	Course Coordinator	Dr. Fawaz Alazemi
Compulsory	No	Prerequisite	0418-223

Catalog Description

The fundamental concept of cloud computing, cloud computing models, cloud storage models, cloud systems management, scalable computing using the cloud.

Textbook

Title	Cloud Computing Solutions Architect
Author	Arshdeep Bahga and Vijay Madisetti
Publisher	VPT
ISBN-10	0996025596

Title	Cloud Computing: Concepts, Technology & Architecture
Author	Ricardo Puttini, Thomas Erl, and Zaigham Mahmood
Publisher	Pearson; 1st edition
ISBN-10	9780133387520

Major Topics Covered in the Course

	Topics	Hours
1	Introduction to Cloud Computing	3
2	Virtual Machines & Compute Services	3
3	Cloud Application Development	9
4	Serverless Applications	6
5	Cloud Storage, Relational and NoSQL databases	6
6	Big Data Analytics in Cloud	6
7	Batch Analytics	6
8	Exams	3
Total		42

Grading

Midterms	30%
Coursework	30%
Final	40%

Learning Outcomes

1. Explain the core concepts of the cloud computing paradigm.
2. Explain and analyze the service models in cloud computing.
3. Explain and analyze security & privacy issues in the cloud.
4. Design & implement a cloud-based application.

Approved by the curricula committee on Jan 25, 2021